A formal knowledge base for amplicon sequencing data and geochemistry of the Rub' al Khali desert

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Abstract

We collected soil and plant material during five campaigns across a 1,043 km transect of the Saudi Rub' al-Khali (Empty Quarter). The resource links 2,516 non-control specimen entries with laboratory processing and a canonical amplicon table of 1,271 profiles, of which 1,237 pass control and read-depth exclusions. Shotgun metagenomes of 551 libraries yield 5,438 species-representative metagenome-assembled genomes (MAGs) with quality estimates, taxonomy, gene annotations and per-library coverage. Associated observations comprise 274 field environmental logs, 712 archived-soil pH measurements, 71 field X-ray fluorescence sessions, 725 laboratory X-ray fluorescence records and 3,234 site-month climate summaries. We represent sites, specimens, processes and measurement values in a modular RDF knowledge base using the Semanticscience Integrated Ontology and validated chemical, environmental and taxonomic references. Shared identifiers preserve specimen identity and assay-specific provenance. Validation checks source reconciliation, measurement admission, identifier mappings, structural constraints and query outputs. The graph supports retrieval across these linked observations through a web portal and SPARQL endpoint. Versioned observations, graph modules and workflow code support comparisons across sites, campaigns and assay types.

Keywords: Knowledge graph, Amplicon sequencing, Shotgun metagenomics, Metagenome-assembled genomes, Rub' al Khali, SIO, Semantic web

1 Background and summary

The Rub' al-Khali (Empty Quarter) is a hyper-arid sand desert that spans approximately 500,000 km^2 of the southern Arabian Peninsula[1]. Its landscape is dominated by extensive linear dunes known as *'uruq* (singular *'irq*, Arabic for "vein"), interspersed with gravel plains, salt flats, and scattered shrub communities[1,2]. Environmental conditions are characterized by large temperature fluctuations, irregular rainfall, and frequent dust storms[3]. Despite these harsh conditions, soil microbial communities also inhabit this extreme environment[4], where spatial variation in soil properties and temporal changes in environmental conditions may influence their composition and distribution. Understanding these communities requires observations that account for both the spatial heterogeneity of the landscape and its changing environmental conditions.

Previous studies of soils in Saudi Arabia have surveyed microbial communities along elevation gradients[5], reponses to rainfall across soil compartments[6], and differences between bulk and rhizosphere soils across regions[7]. A study investigated microbial communities associated with desert plants[8]. Repeated sampling across sites, soil compartments, and time points provides an opportunity to extend this work by examining microbial communities across both spatial and temporal scales. Integrating data at such a scale introduces data management requirements as it becomes necessary to preserve the relationship between sampling locations, collection events, laboratory analyses, and environmental measurements.

Several resources provide models for organizing environmental and microbiome observations. Multi-campaign initiatives, such as Tara Oceans, link campaigns, stations, and sampling events to environmental and sequence data[9]. MIxS/MIMARKS specify contextual metadata for environmental sequencing data[10]. The National Microbiome Data Collaborative promotes the use of persistent identifiers and shared schemas to connect microbiome datasets[11]. Knowledge graphs, including KG-Microbe[12] and MetagenomicKG[13], provide complementary approaches for integrating taxonomic, functional and environmental information. Building on these preinciples, we present a linked representation of field samples, sampling events, laboratory processes, and environmental measurements from a repeated survey of the Rub' al-Khali.

We collected soil samples during five campaigns between 2023 and 2025 along a 1,043 km transect from the west to the east of the Rub' al-Khali in Saudi Arabia. The survey includes 60 primary sites that were repeatedly sampled across campaigns, as well as 10 additional named locations visited in a single campaign. The resulting catalogue contains 2,550 sample records, including 1,271 samples with amplicon sequencing data and 551 samples with shotgun metagenomic sequencing data. From the shotgun data, we reconstructed a catalogue of 5,438 species-representative metagenome-assembled genomes (MAGs). Our resource links the samples to field environmental observations, pH measurements, X-ray fluorescence (XRF) measurements, and climate data.

To improve interoperability and enable integrated queries across heterogeneous data sources, we represent the resource using established ontologies and standardized identifiers. We represnt sites, samples, processes, qualities, and measurement values using the Semanticscience Integrated Ontology (SIO)[14]. This representation explicitly links sequencing results and environmental measurements to the samples and collection events they derive from, retaining their connections to the underlying data-generation processes. We encode chemical entities with ChEBI[15], environmental classifications with ENVO[16], and taxonomic identifiers with NCBI Taxonomy[17]. The resulting linked RDF modules support queries across sampling campaigns, samples, microbial communities, and environmental observations while preserving the context in which each observation was generated.

We performed technical validation to assess the consistency and integrity of the resource, including reconcilation with source data, measurement inclusion, identifier mappings, and structural constraints. We also tested executable queries to demonstrate how information can be retrieved across linked datasets. We report the scope of each validation separately, including the ontology constructs and moducles processed by the ELK reasoner[18]. The portal (https://rubalkhali.science/) provides site-level pages and a SPARQL endpoint for querying the resource. The Data Records section describes the available datasets, and Data Availability specifies their access conditions. A companion manuscript presents analyses of microbial community composition and environmental associations[19], while this Data Descriptor documents the underlying data, their representation, and the provisions supporting their reuse.

2 Methods

2.1 Study description

2.1.1 Study area

We conducted five sampling campaigns along a transect across the Rub' al-Khali starting from the western escarpment (19.15°N, 45.17°E) to the eastern Saudi Arabian border with Oman (20.71°N, 54.73°E). The 60 primary sampling sites span 1,015 km in a straight line and 1,043-km along the transect, calculated by summing great-circle distances between successive sites using an Earth radius of 6,371 km. Figure 1 shows the site coordinates and their altitudes.

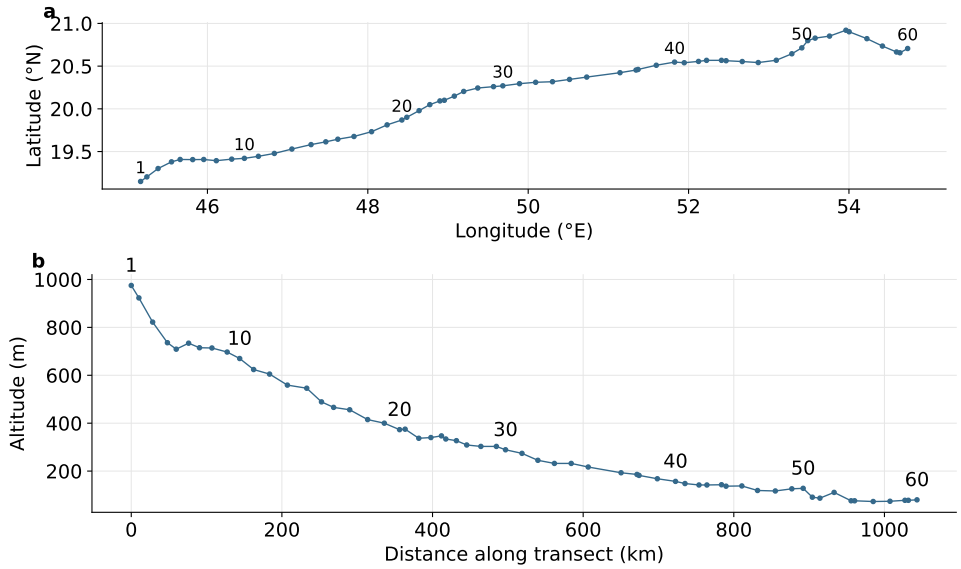


Fig. 1 Primary sampling transect across the Rub' al-Khali. (a) coordinates of sites 1–60. (b) altitude and cumulative great-circle distance.

In addition to the 60 primary sites, the survey included 10 named locations such as wells, a hot spring, and an oil-spill site. These locations were not part of the repeated sampling campaign. Comparisons across campaigns therefore focus on the 60 primary sites, and metadata links each observation to its sampling site, campaign, and sample.

2.1.2 Sampling design

To capture temporal variability, we conducted campaigns across different seasons: late winter 2023 (Trip 1), summer 2023 (Trip 2, truncated to 8 sites due to extreme temperatures), winter 2024 (Trip 3), summer 2024 (Trip 4), and autumn 2025 (Trip 5). Table 2 summarizes sites visited during each campaign.

At each site, we sampled three soil compartments: bulk surface soil, bulk subsurface soil (5–10 cm deep), and subsurface soil collected near roots (5–10 cm deep, hereafter "root-adjacent"). We collected three field replicates per compartment using sterile tubes and sterile plastic spoons, or three tubes taped together. We stored samples at −20°C in the field in an Engel MR040F portable freezer, and transferred samples to an −80°C freezer in the laboratory at the end of each campaign. Table 1 summarizes soil samples collected by campaign and compartment.

Table 1 Soil samples collected and amplicon profiles obtained by campaign and compartment. Control samples are excluded.

	Surface		Deep		Root-adjacent		Total	
Trip	Collected	Profiles	Collected	Profiles	Collected	Profiles	Collected	Profiles
1	192	114	192	110	192	106	576	330
2	24	6	24	8	24	13	72	27
3	180	155	180	141	180	154	540	450
4	180	60	180	59	180	58	540	177
5	180	29	180	55	180	133	540	217
Total	756	364	756	373	756	464	2,268	1,201

2.1.3 Field observations

We recorded air temperature, atmospheric pressure, and relative humidity during site visits, although coverage varied between campaigns (Table 2. Temperature is reported in degrees Celsius, pressure in pascals, and relative humidity in percent. Additionally, field researchers classified each site by an observed environmental feature (*e.g.* sand dune, saline pan, gravel), and we mapped these features to ENVO [16].

Table 2 Campaign overview and field measurement coverage.

Campaign	Season	Sites visited	Field measurements
Trip 1	Late Winter 2023	1–60; 4 named	Temp, Press, Hum.
Trip 2	Summer 2023	1–8	Temp.
Trip 3	Winter 2024	1–60; 5 named	Temp, Press, Hum.
Trip 4	Summer 2024	1–60	Temp, Hum.
Trip 5	Autumn 2025	1–60; 2 named	Temp, Press, Hum.

2.2 Sample processing

2.2.1 DNA extraction

We extracted genomic DNA from 0.9–1.2 g of soil using either the QIAGEN PowerLyzer PowerSoil Kit (catalogue no. 12855-50) or the QIAGEN DNeasy PowerSoil Pro Kit (catalogue no. 47014). The extraction kit and method were recorded for samples when available. Frozen soil samples were thawed on ice and processed immediately. Before homogenization, we incubated samples at 65°C for 10 min with agitation at 1,200 rpm, then homogenized using a PowerLyzer 24 Homogenizer at 4,000 rpm for one 45-second cycle. The remaining purification steps were performed either manually or using a QIAcube Connect instrument, following the manufacturer's protocol. We eluted DNA in 50 µL of the elution buffer supplied with each kit, and stored extracted DNA in −20°C.

We quantified extracted DNA using the Qubit dsDNA High Sensitivity assay, with fluorescence measured using a Varioskan LUX multimode plate reader (Thermo Fisher Scientific). We recorded extract concentrations and processing information for each sample.

2.2.2 16S rRNA amplification

We amplified the V3–V4 region of the bacterial 16S rRNA gene using the Bakt_341F (`CCTACGGGNGGCWGCAG`) and Bakt_785R (`GACTACHVGGGTATCTAATCC`) primers[20], both carrying Illumina overhang adapters. We performed first-round PCR using TaKaRa Ex Taq DNA Polymerase (catalogue no. RR001B). Each 50-µL reaction contained 5 µL of 10× buffer with Mg^{2+}, 4 µL of dNTP mix (2.5 mM each), 2 µL of each primer (10 µM), 0.25 µL of polymerase (5 U/µL), 5 µL of DNA and 31.75 µL of nuclease-free water, giving final concentrations of 0.4 µM of each primer, 0.2 mM of each dNTP and 1.25 U of polymerase. Cycling conditions were 95°C for 3 min; 35 cycles of 95°C for 10 s, 65°C for 30 s and 72°C for 1 min; and a final extension at 75°C for 5 min.

We checked PCR product on a 1% agarose gel for an expected amplicon size of 500 bp. We re-amplified samples without a visible product with 8 µL and then 12 µL of DNA, and we set DNA input by volume because most extracts had low or undetectable concentrations. Samples that failed all three attempts were recorded as failed.

2.2.3 Indexing and sequencing

We added dual indices and Illumina sequencing adapters in a second-round PCR using the QIAGEN Multiplex PCR Kit (catalogue no. 206143), following the Illumina 16S Metagenomic Sequencing Library Preparation protocol[21]: initial activation at 94°C for 15 min; 8 cycles of 94°C for 30 s, 55°C for 30 s and 72°C for 30 s; final extension at 72°C for 5 min.

We purified indexed libraries using Agencourt AMPure XP beads (0.8:1 bead to sample volume), quantified with the Qubit dsDNA High Sensitivity assay, and checked for fragment size on an Agilent 4200 TapeStation (D5000 ScreenTape). We pooled libraries in equimolar amounts, calculating molarity from concentration and fragment size. Pools were sequenced on an Ollumina NovaSeq 6000 with an SP flow cell and a

500-cycle kit (2×250 bp paired-end reads). Library preparation was performed in the PI's laboratory and sequencing at the KAUST Bioscience Core Lab.

2.2.4 Control samples

We followed recommended guidelines for reporting contamination in low biomass samples by having controls for differnent study stages[22] (Table 3). To control for contamination introduced during DNA extraction, we included extraction blanks with nuclease-free water as input instead of soil, and processed them the same way as soil samples. We included an extraction blank for each extraction kit. Coverage varied by extraction batch, and we sequenced nine extraction blanks linked to Trip 4 samples, and 17 extraction blanks linked to Trip 5 samples.

We included two types of PCR negative controls. No-template controls (NTCs) contained PCR reagents and nuclease-free water instead of DNA, and reagent-only controls per amplification plate that contained reagents but neither water nor DNA. Reagent-only controls had low or undetectable DNA concentrations and TapeStation profiles of only adapter dimers, so they were not sequenced. We sequenced six NTCs that were prepared with Trip 5 samples.

We also included positive standards to assess how well expected genera were recovered. During Trips 1 and 2 we used the ZymoBIOMICS HMW DNA Standard (D6322), which contained purified DNA and was added at the PCR step. D6322 contains seven bacterial genomes (14% of genomic DNA each) and one yeast (2%). During Trip 3 we used the ZymoBIOMICS Microbial Community Standard (D3600) which contains whole cells and was therefore included from the extraction step. D6300 contains eight bacteria (12% of genomic DNA each) and two yeasts (2% each). These proportions are the manufacturer's values. Two of each type of positive standard were amplified and sequenced.

Table 3 Laboratory controls by campaign.

Trip	Extraction blanks	PCR negative controls	Positive standards
1	One per kit; PowerSoil Pro blank sequenced	Nuclease-free water	
2	One per kit	NTC-2	Two D6322 (DNA)
3	One per kit	PCRCtrl (index 728)	Two D6300 (whole cells)
4	One per batch; 6 of 9 sequenced	Not sequenced	Checked by gel and TapeStation only
5	One per batch; 18 sequenced	Six NTCs sequenced	D6300, shotgun sequenced

2.2.5 PMA-treated samples

DNA extracted from soil includes DNA from intact cells as well as extracellular DNA and DNA from dead or damaged cells[23]. Propidium monoazide (PMA) is a dye that

enters only cells with damaged membranes and binds to free DNA, and prevents amplification by PCR when exposed to light. We compare PMA-treated and untreated aliquots of the same soil samples to estimate how much of the 16S rRNA gene signal comes from cells with intact membranes. We applied this treatment to a subset of Trip 5 samples.

The PMA dataset includes three soil samples collected from campsites during Trip 5. We collected root-adjacent soil from campsite C1, and root-adjacent and surface soil from campsite C2. From each collection tube, we took three samples and suspended each in phosphate-buffered saline (PBS, pH 7.4) at a soil-to-buffer ratio of 1:10 (w/v). The three samples from each collection tube are technical replicates of the treament and not independent field samples. We split each sample into a treated aliquot that included 50 µM PMAxx, and an untreated aliquot.

We subjected both aliquots to the same procedure. We kept the aliquots in the dark for 10 min to allow the dye to enter the cells, and then exposed them to 450 nm light for 35 min at a distance of 15–20 cm from the light source, and turned them every 15 min to ensure even exposure. We stored aliquots at −20°C, and extracted DNA from 860 µL of each aliquots with the PowerSoil Pro kit.

2.2.6 Data processing

We processed raw amplicon reads with nf-core/ampliseq (v2.14.0)[24,25], using DADA2[26] for error correction and denoising, and SILVA (v138.2)[27] for taxonomic assignment. This resulted in 351,472 amplicon sequence variants (ASVs), and included 1,247 biological samples and 24 control samples. The container digest and full denoising parameters are included in the repository.

We predicted gene family and pathway abundances from the 16S rRNA data with PICRUSt2 v2.4.1[28]. KEGG Orthology (KO) prediction used the phylogenetic independent contrasts method, edge exponent 0.5, and random seed 100. Enzyme Comission (EC) and KO abundances required 100 reads, and pathway abundances were inferred from the EC predictions.

2.3 Soil Measurements

2.3.1 Soil composition (XRF)

We measured soil elemental composition by X-ray fluorescence (XRF) using an Olympus Vanta VMR-XXX-G3-E handheld analyser (serial no. 841307), both in the field during Trip 5 and in the laborator on archived soil samples from all campaigns. For field measurements, we used the instrument in Geochem (3-Beam) mode with NORMAL settings, and retained 71 complete measurement sessions from 58 sites, including repeated sessions at nine sites. In the laboratory, we analyzed 725 samples with the instrument in Fast Mode with the Fast Screening-He8mm setting. Elemental concentrations and their oxide equivalents are recorded as separate variables and are not asssigned units.

To assess agreement between field and laboratory measurements, we compared the median of repeated field sessions at each site with the corresponding Trip 5 laboratory measurements at the 58 sites measured using both methods. This comparison

assesses agreement in rankings rather than absolute concentrations, and we report the correlations in Technical Validation.

We cross-referenced the 93 reported elements against ChEBI r247 (3 December 2025) and PubChem properties (retrieved 23 July 2026) [15,29]. We identified corresponding ChEBI entries for 79 variables and PubChem entries for 91 entries. We verified identifier validity, names, chemical formulae, charge, and whether each variable represented an element or an oxide.

2.3.2 Soil pH

We measured the pH of archived soil samples from all five campaigns. We sieved dried soil to below 2 mm and mixed 1 g of soil with 2.5 mL of 0.01 M $CaCl_2$. We mixed each suspension for 30 min in a HulaMixer and let it stand for another 30 min at room temperature before measuring pH.

We measured pH using an accumet AB315 meter with a Fisher Scientific electrode (cat. no. 13-620-112) which we calibrated at the start of each session, and we recorded the electrode slop and the reading of the pH 10 buffer for every session. We rinsed the electrode with Milli-Q water between measurements. We oretained 712 measurements with a reocrded electrode slop between 95% and 102% and a pH 10 buffer reading within 0.1 units of its nominal value (Table 4).

Table 4 Soil pH measurement coverage.

Campaign	Surface	Subsurface	Root-adjacent	Total
1	52	56	49	157
2	5	6	12	23
3	54	50	52	156
4	50	52	54	156
5	30	57	133	220
Total	191	221	300	712

2.4 Climate data

We retrieved historical climate data for each sampling site from the Open-Meteo archive API[30] using geographic coordinates. We obtained both monthly and daily climate observations. The monthly dataset contains 12,936 values that cover temperature, total precipitation, rainfall, and mean relative humiditiy from January 2022 to January 2026. The daily dataset covers 1 January 2022 to 1 February 2026 and covers temperature, total precipitation, and rainfall.

We also retrieved daily prepipitation data from NASA POWER PRECTOTCORR for sites 1–60 from 1 January 2022 to 31 December 2025, which yielded 87,660 records.

2.5 Shotgun metagenomics and genome catalogue

Shotgun metagenomic sequencing complements the amplicon profiles with genome assemblies and gene annotations from the same sampling campaigns. The read-bearing library inventory contains 551 libraries across 61 site codes: 210 from Trip 1, 37 from Trip 2, 128 from Trip 3, 85 from Trip 4 and 91 from Trip 5. Library metadata records the campaign, source site, soil compartment and replicate. The libraries were sequenced as 554 paired-end runs, 221 on an Illumina NovaSeq 6000 and 333 on an Illumina NovaSeq X; three libraries were sequenced on both instruments. Libraries for 81 runs were prepared with the NEBNext Ultra II DNA Library Prep Kit and for 473 runs with the QIAseq FX DNA Library UDI Kit. The shotgun study is registered at ENA under `PRJEB125153` (secondary accession `ERP204894`).

Genome reconstruction combined MEGAHIT v1.2.9 assembly[31] with single-sample and self-supervised SemiBin2 v2.2.1 binning[32], followed by DAS Tool v1.1.7 refinement[33]. MEGAHIT produced contigs of at least 500 bp for 549 of the 551 libraries; libraries 51Sr1 and F5Dr1 yielded no contigs. CheckM2 v1.1.0[34] estimated completeness and contamination for the refined bins, and 12,434 refined bins reached at least 50% completeness with at most 10% contamination. We dereplicated the bins with dRep[35], using CheckM2 quality estimates, a minimum completeness of 50%, a maximum contamination of 10%, and a minimum genome length of 50,000 bp. The recorded clustering parameters specify 90% primary ANI and 95% secondary ANI, with fastANI for secondary comparisons and a minimum alignment coverage of 0.1 using dRep's `larger` coverage mode. The resulting catalogue contains 5,438 representative identifiers. The provenance table links each identifier to its source library, binning strategy and bin, and records quality estimates from the refined-bin CheckM2 report.

Taxonomic assignments for the refined representatives used GTDB-Tk v2.7.2 with GTDB release R232[36]. The two native classification tables contain 4,847 bacterial and 591 archaeal representatives, and their genome identifiers match the representative catalogue exactly. We detected rRNA genes with barrnap v0.9 and tRNA genes with tRNAscan-SE v2.0.12[37], and assigned each representative a quality tier following the MIMAG standard[38]. Of the 5,438 representatives, 126 meet the high-quality criteria, which require more than 90% completeness, less than 5% contamination, the 23S, 16S and 5S rRNA genes and tRNAs for at least 18 of the 20 amino acids. Another 5,311 meet the medium-quality criteria of at least 50% completeness and less than 10% contamination. The remaining representative, `48Dr2__self_SemiBin_25`, has 10% contamination.

We annotated the genes of each representative with Bakta v1.12.0[39] and its full database v6.0. Bakta predicts coding sequences, rRNA and tRNA genes and translates the coding sequences into proteins. eggNOG-mapper v2.1.13[40] then searched these proteins against the eggNOG v5.0.2 database with DIAMOND v2.0.15 and assigned KEGG orthologs (KOs), CAZy families, Pfam domains and preferred gene names. For each genome, we counted the predicted proteins, the proteins with an eggNOG assignment and the proteins with at least one KO. We also counted, for each genome, the proteins that carry each KO, CAZy family, Pfam domain and preferred name. These feature tables list only the genome–feature pairs that occur in the annotations.

To estimate the coverage of each representative in each library, we mapped the reads of all 551 libraries to a single reference that contains every contig of the 5,438 representatives. We prefixed each contig identifier with its genome identifier, so that contigs with the same assembler-assigned name in different genomes remain distinct. Bowtie2 v2.5.1[41] aligned the first 200 million read pairs of each library in `--very-sensitive` mode and retained aligned reads only; SAMtools v1.16.1[42] sorted the alignments. CoverM v0.6.1[43] in genome mode counted reads aligned with at least 95% identity and reported, for every genome, the mean depth, the covered fraction, the trimmed mean depth, the number of mapped reads and the genome length.

The ecology comparison uses a separate 991-genome catalogue and 150 library profiles from Trips 1–3 (58, 15 and 77 libraries, respectively). Its CoverM profiles[43] report genome-relative abundance, and eggNOG-mapper v2.1.12[40] annotations yield a 990-genome KO copy-number matrix after intersecting the annotated genomes with the abundance profiles. These inputs and the six source tables for the MAG-derived versus PICRUSt2 comparison retain their analytical-subset provenance.

2.6 Knowledge representation

We present the resource as a modular RDF knowledge base that connects sampling sites, physical samples, laboratory measurements, experimental producedures, sequencing datasets, and taxonomic assignments. Rather than representing these data as independent tables, we encode their relationships explicitly. This allows users to trace an observation to the sample, site, or process it describes, and query related data across different stages of the study (Figure 2).

The graph separates a shared terminology module, which defines the classes and relationships used to describe the data, from modules containing the actual study records. In ontology terminology, these correspond to the Tbox and ABox, respectively. Shared identifiers connect the modules (Table 7), while the Semanticscience Integrated Ontology (SIO) provides general classes and relationships for representing entities, processes, and measurements[14]. The terminology module declares 333 classes (297 defined for this project and 36 from other ontologies), 20 object properties, and 35 datatype properties written in OWL[44].

To support interpretation and reuse, KG v3.0.1 provides English definitions for all 342 project terms, recorded using SKOS and IAO definition annotations. Claude Sonnet 5.5 drafted the definitions from term labels, axioms and dataset context, and OpenAI Codex assisted with automated review and editing. We distribute the definitions, model provenance and review records with the release. The terminology module imports four reference ontologies, and the observation modules are linked as parts of the dataset.

2.6.1 Representing provenance

The data model rests on two distinctions. First, physical materia (a soil sample, a DNA extract), the process applied to it (extraction, sequencing), and the data those processes produce (a FASTQ dataset) are represented as separate entities connected by inputs and outputs. Second, a measured value is kept separate from the property

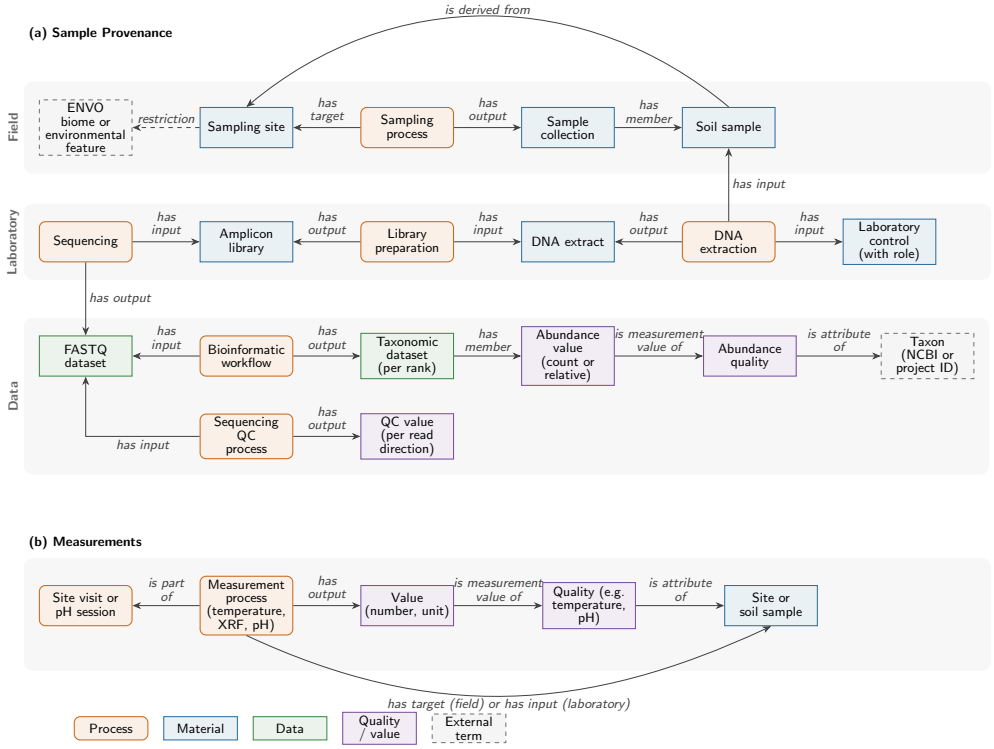


Fig. 2 Main patterns in the knowledge graph. (a) A sampling process targets a site and produces a collection of soil samples, each also linked directly to its site. Laboratory processes are connected through their inputs and outputs, from DNA extraction to sequencing; laboratory controls enter at the processing step whose role they test. A bioinformatic workflow converts each FASTQ dataset into taxonomic datasets at seven ranks, whose values quantify the abundance of taxa identified by NCBI Taxonomy or project identifiers. Sites are linked to ENVO classes through existential restrictions (dashed). (b) Each measurement process produces a value that quantifies a quality of a site or soil sample, and is part of a site visit or pH measurement session. Arrows point from subject to object of each relation. Protocols linked to each process are omitted for clarity.

it quantifies and from the entity that has the property. Together, these distinctions record where every value came from.

Shared vocabularies and external ontologies. We reuse established terms that exist in an external ontology where relevant (Table 5). Because these identifiers are shared with other resources, the data can be queried and compared alongside other datasets. We retain original labels and values with the mapped identifiers.

Measurements. We represent measurements separately from the biological or environmental qualities they describe. For example, the temperature of a site is a property (quality) of that site (Listing 1). A measurement process carried out during a particular site visit produces a value for that property. The graph links the value to the quality, the quality to the site, and the measurement process to the visit. This separates what was measured from how, where, and when it was measured. It also

Table 5 External ontologies and reference resources used in the knowledge graph.

Resource	Used to describe
SIO [14]	Processes, inputs and outputs, values, qualities and the relations between them
PATO	Measured qualities
UO [45]	Units of measurement
ENVO [16]	Biomes and environmental features of sites
NCBI Taxonomy [17]	Taxa
ChEBI [15], PubChem [29]	Elements and oxides reported by XRF

allows the same quality to be measured repeatedly, for example on different visits, without ambiguity.

Each value holds its numerical reading and, where documented, its unit. The ShEx shape assigned to each type of value specifies its datatype and unit requirements, which differ for counts, measurements with units and measurements whose units were not documented.

Listing 1 Field temperature and its visit provenance.

```
@prefix rak: <https://rubalkhali.science/kb/RAK_> .
@prefix sio: <http://semanticscience.org/resource/SIO_> .
@prefix pato: <http://purl.obolibrary.org/obo/PATO_> .
@prefix uo: <http://purl.obolibrary.org/obo/UO_> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

# Measurement process
rak:P000001 a rak:0000006 ; # type: field measurement process
    sio:000068 rak:3000001 ; # is part of -> site visit
    sio:000229 rak:4000001 . # has output -> measured value

# Site visit
rak:3000001 a rak:0000003 ; # type: site visit
    sio:000291 rak:1000001 . # has target -> sampling site

# Measured value
rak:4000001 a rak:0000010 ; # type: measurement value
    sio:000221 uo:0000027 ; # has unit -> degree Celsius
    rak:2000003 "20.7"^^xsd:double ; # numerical value
    sio:000215 rak:5000001 . # is measurement value of -> quality

# Quality of the site
rak:5000001 a pato:0000146 ; # type: temperature (PATO)
    sio:000011 rak:1000001 ; # is attribute of -> sampling site
    sio:000216 rak:4000001 . # has measurement value -> value
```

Taxonomy. A taxonomic label can denote different taxa across databases and releases, and a taxon can be referred to by several labels [46]. To ensure consistent

identification, we retained SILVA assignments for each ASV and linked it to either a validated NCBI Taxonomy identifier or a project-specific identifier when no suitable NCBI Taxonomy identifier was found.

Each SILVA assignment describes the organism across seven ranks from domain to species, and may include an additional species prediction and a confidence score. We used the species name from the rank assignment in the knowledge graph and kept the additional species assignment as an annotation. Before mapping, we removed prefixes from taxon names and explicitly recorded missing and unclassified ranks.

For each SILVA assignment, we checked for a candidate NCBI identifier from a fixed NCBI Taxonomy release (OBO Foundry NCBITaxon r2025-03-13 built from NCBI Taxonomy 1 March 2025). A candidate was only accepted if the identifier exists in that release, its rank matches, its name matches the SILVA name or a documented synonym, and its parent taxa in NCBI match identifiers already accepted for the higher ranks of the same assignment. When no candidate met these criteria, the name was given a project-specific identifier, which refers to a unique combination of rank, name, and parent taxa. As a result, users can therefore query each ASV by its original SILVA assignment or by its NCBI or project identifier.

2.6.2 Formal patterns

The relationships described above are defined formally as Description Logic patterns[47]. Each pattern specifies which kinds of entities can be related and how, giving a precise description of the graph's structure that users can rely on when writing queries. The same patterns define the constraints checked during structural validation (Section 2.6.5). In the notation below, $\sqsubseteq$ means "is a subclass of", $\sqcap$ means "and", $\exists R.C$ means "has at least one relation R to a member of class C", and $\forall R.C$ means "all relations R point to members of class C". Section 2.6.3 gives the corresponding RDF examples.

Sampling. A sampling process targets a site and produces a collection of samples, and each sample is also linked directly to its site.

$$\text{SamplingProcess} \sqsubseteq \text{Process} \tag{1}$$

$$\text{Sample} \sqsubseteq \text{MaterialEntity} \tag{2}$$

$$\text{SampleCollection} \sqsubseteq \text{Collection} \sqcap \forall \text{hasMember.Sample} \tag{3}$$

$$\text{SamplingProcess} \sqsubseteq \exists \text{hasTarget.Site} \sqcap \exists \text{hasOutput.SampleCollection} \tag{4}$$

$$\text{Sample} \sqsubseteq \exists \text{isDerivedFrom.Site} \tag{5}$$

Protocols. Each field or laboratory step is recorded as the execution of a protocol, with the material or data it used and produced.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \tag{6}$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \tag{7}$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy.Protocol} \tag{8}$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput.Entity} \sqcap \exists \text{hasOutput.Entity} \tag{9}$$

Taxonomic abundance. A value quantifies the abundance of a taxon and belongs to a dataset produced by a bioinformatic workflow from a sequence dataset.

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \tag{10}$$
$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \tag{11}$$
$$\text{BioinformaticWorkflow} \sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset} \tag{12}$$
$$\text{TaxonomicDataset} \sqsubseteq \forall \text{hasMember.AbundanceObservation} \tag{13}$$
$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AbundanceQuality} \tag{14}$$
$$\text{AbundanceQuality} \sqsubseteq \exists \text{isAttributeOf.Taxon} \tag{15}$$
$$\text{AbundanceQuality} \sqsubseteq \exists \text{hasMeasurementValue.AbundanceObservation} \tag{16}$$

XRF. Laboratory XRF measurements take a soil sample as input, whereas field measurements target a sampling site, and both produce concentration values.

$$\text{XRFProcess} \sqsubseteq \text{MeasurementProcess} \sqcap \exists \text{hasOutput.Concentration} \tag{17}$$
$$\text{LabXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasInput.SoilSample} \tag{18}$$
$$\text{FieldXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasTarget.SamplingSite} \tag{19}$$
$$\text{Concentration} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AnalyteQuality} \tag{20}$$

Sequencing. DNA extraction, library preparation and sequencing are linked through their shared inputs and outputs, so that each sequence dataset can be traced back to its soil sample.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \tag{21}$$
$$\text{DNAExtractionProcess} \sqsubseteq \exists \text{hasInput.SoilSample} \sqcap \exists \text{hasOutput.DNAExtract} \tag{22}$$
$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \tag{23}$$
$$\text{LibraryPreparationProcess} \sqsubseteq \exists \text{hasInput.DNAExtract} \sqcap \exists \text{hasOutput.AmpliconLibrary} \tag{24}$$
$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \tag{25}$$
$$\text{SequencingProcess} \sqsubseteq \exists \text{hasInput.AmpliconLibrary} \sqcap \exists \text{hasOutput.FASTQDataset} \tag{26}$$
$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \tag{27}$$
$$\text{FASTQDataset} \sqsubseteq \forall \text{hasMember.SequenceRead} \tag{28}$$

Sequence quality control. Quality metrics are attached to the FASTQ dataset they describe.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \tag{29}$$
$$\text{SequencingQCProcess} \sqsubseteq \exists \text{hasInput.FASTQDataset} \sqcap \exists \text{hasOutput.QCMeasurementValue} \tag{30}$$
$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.QCQuality} \tag{31}$$
$$\text{QCQuality} \sqsubseteq \exists \text{isAttributeOf.FASTQDataset} \tag{32}$$

2.6.3 Graph examples

The following examples instantiate the formal patterns with observations from the resource. Table 6 defines the namespaces used in the Turtle examples and SPARQL queries. These declarations must be prepended when parsing an example. The listing validator compares each example with the corresponding modules and pinned reference terms.

Table 6 Prefix declarations used in all Turtle and SPARQL listings.

Prefix	Namespace
rak:	https://rubalkhali.science/kb/RAK_
rak_kb:	https://rubalkhali.science/kb/
sio:	http://semanticscience.org/resource/SIO_
owl:	http://www.w3.org/2002/07/owl#
pato:	http://purl.obolibrary.org/obo/PATO_
uo:	http://purl.obolibrary.org/obo/UO_
envo:	http://purl.obolibrary.org/obo/ENVO_
ncbitaxon:	http://purl.obolibrary.org/obo/NCBITaxon_
obo:	http://purl.obolibrary.org/obo/
rdf:	http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs:	http://www.w3.org/2000/01/rdf-schema#
xsd:	http://www.w3.org/2001/XMLSchema#
geo:	http://www.opengis.net/ont/geosparql#
dcterms:	http://purl.org/dc/terms/

Listing 2 represents the Trip 1, Site 1 shallow-subsurface sample. The sampling process targets a site and outputs a collection containing the samples. The sample also links directly to its source site, allowing queries to recover provenance without traversing the sampling event.

Listing 2 Turtle representation of a sampling process.

```
rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:7000001 a sio:001419 ; # Material sample collection
  sio:000059 rak:6000001 . # hasMember

rak:6000001 a rak:0000021 ; # Deep Soil Sample
  sio:000244 rak:1000001 . # isDerivedFrom
```

The same sample is the input of the extraction process in Listing 3. A protocol specification identifies the procedure, and the output DNA extract carries a concentration quality. A separate measurement value quantifies that quality through the reciprocal SIO measurement relations.

Listing 3 Turtle representation of a DNA extraction protocol.

```
rak:P200001 a sio:000994 ; # Scientific Protocol Execution
  sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
  sio:000008 rak:5900001 . # hasAttribute (Concentration Quality)
```

Taxonomic observations bind a count to both a taxon and its source sequence dataset (Listing 4). Here, sequencing run `ERR16783308` from sample `V10Dr1` has 853,814 reads assigned to Bacteria. Each taxonomic dataset contains one rank and one abundance basis. Count and relative abundance datasets retain the same classifier lineage and source sample context.

Listing 4 Turtle representation of taxonomic abundance formalized from a feature table.

```
rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000339 rak:L000012 ; # 16S amplicon processing protocol
  sio:000230 rak:7791009 ; # hasInput (FASTQ Dataset, ERR16783308)
  sio:000229 rak:7740001 . # hasOutput (Taxonomic Dataset)

rak:7740001 a rak:0000074 ; # taxon absolute abundance dataset
  sio:000059 rak:V00000001 . # hasMember (Observation Value)

rak:V00000001 a rak:0000076 ; # Absolute abundance value
  sio:000215 rak:Q00000001 ; # isMeasurementValueOf
  rak:2000025 "Domain: Bacteria" ; # classifier lineage
  rak:2000026 "853814.0"^^xsd:double . # absolute read count

rak:Q00000001 a rak:0000078, pato:0000070 ; # Absolute Abundance Quality
  sio:000011 ncbitaxon:2, rak:7791009 ; # taxon and FASTQ bearers
  sio:000216 rak:V00000001 . # hasMeasurementValue
```

Laboratory XRF specializes the measurement pattern by linking the process to the archived sample from which it was measured. Listing 5 shows the aluminium observation for `V1Sr1`. Its numerical value and quality have reciprocal links. The value retains the instrument reported magnitude with undocumented concentration units. Field sessions instead identify a target site through `sio:000291`.

Listing 5 Turtle representation of an XRF geochemistry measurement.

```
rak:P300072 a rak:0000025 ; # XRF Measuring Process
  sio:000230 rak:6001732 ; # hasInput (sample V1Sr1)
  sio:000339 rak:L_Surface_Lab ; # laboratory protocol
  sio:000229 rak:4302121 . # hasOutput

rak:4302121 a rak:0000507 ; # Al concentration value
  sio:000215 rak:5302121 ; # isMeasurementValueOf
  rak:2000012 "0.78"^^xsd:double . # Concentration

rak:5302121 a rak:0000107 ; # Al concentration quality
  sio:000011 rak:6001732 ; # isAttributeOf
  sio:000216 rak:4302121 . # hasMeasurementValue
```

Shared inputs and outputs connect extraction, library preparation and sequencing (Listing 6). The final sequencing dataset links to its ENA accession. A SPARQL traversal can therefore recover the recorded provenance from sample to sequencing run.

Listing 6 Turtle representation of the sequencing workflow from soil sample T1Dr1 to ENA run ERR16061083.

```
rak:P200923 a sio:000994 ; # DNA Extraction Process
  sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000577 ; # hasInput (T1Dr1 Soil Sample)
  sio:000229 rak:6800540 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000065 ; # Library Preparation Process
  sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
  sio:000230 rak:6800540 ; # hasInput (DNA Extract)
  sio:000229 rak:6690001 . # hasOutput (Amplicon Library)

rak:P280001 a rak:0000066 ; # Sequencing Process
  sio:000230 rak:6690001 ; # hasInput (Amplicon Library)
  sio:000229 rak:7790001 . # hasOutput (FASTQ Dataset)

rak:7790001 a rak:0000063 ; # FASTQ Dataset
  rdfs:seeAlso <https://identifiers.org/insdc.run/ERR16061083> .
```

Quality control observations retain the read direction through separate forward and reverse metric classes. Listing 7 gives the forward-read GC content for run ERR16062319: 58.0%. The quality belongs to the same FASTQ dataset consumed by the quality control process.

Listing 7 Turtle representation of forward-read GC content for ENA run ERR16062319.

```
rak:P350001 a rak:0000200 ; # Sequencing QC Process
  sio:000230 rak:7790263 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:4450001 . # hasOutput (QC Value)

rak:4450001 a rak:0000217 ; # Forward GC Content Value
  sio:000215 rak:5550001 ; # isMeasurementValueOf
  sio:000221 uo:0000187 ; # Percent
  rak:2000074 "58.0"^^xsd:double . # has GC content

rak:5550001 a rak:0000205 ; # Forward GC Content Quality
  sio:000011 rak:7790263 ; # isAttributeOf (FASTQ Dataset)
  sio:000216 rak:4450001 . # hasMeasurementValue
```

2.6.4 OWL and RDF implementation

The modules are generated with Groovy scripts using OWL API v5.1.20[48]. Keeping the shared terminology separate from the study data allows each to be updated and checked independently[49]. The asserted graph is stored in OpenLink Virtuoso[50] which supports SPARQL queries over the whole resource. Release KG v3.0.1 serves the asserted triples.

Some facts follow from the asserted statements without being stated explicitly; for example, a sample of a subclass of soil sample is also a soil sample. A separate workflow derives such facts with a rule program v1.5.0 of positive RDF rules and stores them in a separate graph, so that recorded statements can be distinguished from those that are derived. The rules cover equivalent classes and properties, subclass and subproperty hierarchies, instance types, property domains and resource-valued ranges, and inverse, symmetric, and transitive properties. For property chains, one rule is generated for each non-empty finite chain in the loaded schema, including chains

with inverse properties. Further rules handle value restrictions, existential restrictions (classified from existing links and typed targets), universal restrictions (applied to existing targets), intersections, union membership and enumerated classes. The rules do not create new RDF terms. The schema audit lists the imported ontologies and the OWL constructs not covered by the rules.

Each rule adds only triples that are absent from both the asserted and the inferred graph, working in batches, and is repeated until it adds no new triples. The complete set of rules is applied repeatedly until a full pass adds no triples; at that point, all consequences of the rule program have been computed (its least fixed point). The run records the rule text, the lists of data partitions, the number of inferred triples, software versions and input identifiers. It stops with a failure if a query result is truncated, an SQL error occurs or the maximum number of passes is reached. After the run completes, separate checks search for conflicts between disjoint or complementary classes, members of `owl:Nothing` and incompatible control roles. The inference therefore covers only this rule set and these checks.

2.6.5 Validation

We checked the generated graph for structural integrity with Shape Expressions (ShEx) and for logical consistency with an OWL reasoner. Both checks detect errors in how the data are represented, but they do not establish biological correctness. Results are reported in Section 4.

Structure. ShEx[51] is a schema language that checks whether selected entities have the required types and properties, how many values each property has and whether the values have the correct datatype. We defined shapes for all main entity types: sampling sites, soil samples, DNA extracts, sequencing datasets, measurements, quality-control records, laboratory control materials, control roles, laboratory batches, sequences detected in controls, soil pH measurements, values and sessions, and taxonomy mappings. For example, the sample shape requires a replicate type, a label, a source identifier and a link to the site from which the sample was derived, and the DNA concentration shape checks the numerical value, the unit (ng/µL) and the links to the measured quality and process.

The shapes have known limits. The site shape checks that a geometry is present, but not that it is valid WKT with correct coordinate syntax. The XRF-process shape requires an input or target, but checking that it belongs to the correct class requires an additional check. The shapes are open, so entities may have properties beyond those declared, and additional types are accepted regardless of how they were asserted. Test data that deliberately violate the shapes confirm that the failures reported in Technical Validation are detected.

A Shape Map assigns each set of entities (e.g. all sampling sites) to its shape. During deployment, each newly generated data module is checked, and deployment stops if a required property is missing (e.g. the coordinates of a site) or a value has the wrong datatype. Validation used the Apache Jena ShEx library v4.10.0, called from Groovy scripts. The multi-gigabyte taxonomy data are too large for this in-memory check; instead, the file is parsed in full with an independent Turtle parser and checked with

a streaming validator that covers every link between classification process, dataset, quality and value.

Logical consistency. We checked logical consistency with the ELK reasoner v0.4.3[18] through OWL API v4.5.26, after merging the modules that ELK can process into a single ontology. ELK supports only part of the OWL language, and the modules also contain universal restrictions, inverse properties and other constructs that ELK does not handle. The result therefore applies only to the supported subset; an OWL profile report and a checked projection of the ontology would be needed to define its logical scope precisely. The taxonomy data and large reference modules were excluded.

2.7 Reproducible workflow

A Nextflow[52] workflow generates the knowledge graph modules, runs the validation checks, and computes the statistics reported in this manuscript. Configuration profiles are available for running the workflow locally or a on a computing cluster. Sample processing (Section 2) was run seperately with its own parameters and software environments.

The workflow produces each versioned release of the knowledge graph by generating the data modules, comparing their checksums with those listed in the release manifest, loading them into a new database, and checking the results of test queries against the source data. The public service is switched to the new release only after these checks pass. The lease manifests list the fixed input files and the generated modules along with with their checksums.

3 Data records

KG v3.0.1 contains 45,707,660 asserted triples from 16 modules and is archived at Zenodo[53]. The resource contains the modules listed in Table 7, the combined asserted graph, and a reproducibility package with code, term definitions and validation evidence. The modules use RDF/XML and Turtle, and the combined graph uses compressed N-Quads. Shared site, sample and process identifiers connect observations across modules. The deposit also includes an inventory with checksums.

The accompanying data repository described in Data Availability contains:

- A sample catalogue linking collection visits, samples, extractions, libraries and sequence runs.
- Amplicon counts and taxonomic mappings, field environmental observations, daily climate inputs and monthly climate summaries.
- Laboratory and field XRF values, the chemical mapping table for 93 reporting channels, and archived-soil pH measurements (`EQ-PH-SHARED-v1.0.1`).
- The refined MAG catalogue: 5,438 genome FASTAs with per-genome checksums, source-library and binning provenance, CheckM2 quality estimates and GTDB R232 classifications, together with MIMAG quality tiers and the ENA accessions of each library, assembly, bin and MAG. Shotgun library metadata covers 551 read-bearing libraries (Section 2.5).

Table 7 The 16 RDF modules included in KG v3.0.1.

Module	Content
Project terminology	Base ontology (TBox)
Sampling sites	Sampling sites with coordinates and biome annotations
Samples	Soil and plant sample individuals with compartment types
Environmental observations	Field environmental measurements and climate records
XRF measurements	XRF measurement values from the field and laboratory workflows
DNA extracts	DNA extracts with concentration data
Sequencing runs	Sequencing run records linked to ENA accessions
Sequence quality	Sequencing quality metrics (GC content, duplicate rate, sequence counts)
Laboratory controls	Laboratory-control materials, roles, batches and sequence occurrences
Soil pH	Archived-soil pH measurements, sessions and specimen links
Taxonomic observations	Taxon observations over feature profiles
Taxonomic vocabulary	Vocabulary supporting canonical taxonomic mappings
SIO	Semanticscience Integrated Ontology v1.59
ENVO	Environment Ontology, 20 October 2025
PATO	Phenotype and Trait Ontology, 14 May 2025
Units Ontology	Units Ontology, 25 May 2023

- Functional annotations of the refined MAG catalogue: per-genome counts of KOs, CAZy families, Pfam domains and preferred gene names, per-genome numbers of predicted and annotated proteins, and the tool versions and checksums of each genome's annotation.
- Coverage profiles of the 551 shotgun libraries against the refined MAG catalogue, with the mapping reference definition that assigns each contig to its genome.
- The paired PMA profiles and the ecology-analysis inputs: 150 CoverM profiles from Trips 1–3 against the 991-genome catalogue, eggNOG annotations supporting its 990-genome analytical subset, and six source tables for the MAG-derived versus PICRUSt2 comparison.
- A data dictionary defining fields, data types, units and missing-value conventions; a climate inventory with acquisition metadata; and a file inventory with sizes, checksums, licensing status and bulk-data retrieval information.
- Validation schemas, SPARQL queries, expected query outputs and source comparisons.

Tabular data use delimited text formats while larger collections are compressed.

4 Technical validation

We validated the resource for agreement between the released data and the source records, including checks on laboratory controls and agreement between measurements. We also validated the structure, logical consistency and integrity of the knowledge graph. Additionally, we checked whether queries on the graph return expected results. For every check, the validation records identify input files by checksum and the software versions used (Section 3).

4.1 Agreement with source records

Every eligible record in the source tables is represented once in the graph. This includes 2,516 samples, 1,647 DNA extracts, and 1,242 amplicon libraries and their corresponding FASTQ datasets. All 274 field environmental measurements for temperature, pressure, and relative humidity pass checks for site identity and for plausible ranges. All 93 reported XRF elements were checked against ChEBI and PubChem, and elements without mapped identifiers are included with an empty mapping field. All 1,168 pH records and their acceptance or exclusion conditions were reproduced, and accepted measurements were linked to the correct samples.

4.1.1 Laboratory controls and measurement agreement

The PowerSoil Pro extraction-control profile contained 1,165 ASVs, with 84% of reads assigned to ASVs shared with biological profiles. Its nearest canonical profile was 46Dr1, with Bray–Curtis dissimilarity 0.39. We excluded this control from the contaminant screen. A library-order test compared nearby library numbers with different-site pairs matched by geographic-distance decile and compartment, using 999 library-order permutations. The Trip 1 and Trip 3 Bray–Curtis contrasts were -0.003 ($p = 0.15$) and $+0.006$ ($p = 0.99$), respectively. These contrasts summarize the tested preparation-order association.

To assess agreement between field and laboratory XRF measurements, repeated field sessions were reduced to site medians and paired with the Trip 5 subsurface laboratory observation at 58 sites. Among 12 analytes with positive values in both workflows at more than 10 paired sites, the median Spearman correlation was 0.21 (range -0.40 to 0.48). This diagnostic describes site-level rank agreement among the retained positive pairs.

4.2 Competency questions and SPARQL queries

Competency questions define the intended retrieval tasks (Table 8). The archived query workload covers site, geochemical, molecular-provenance and taxonomic data. The Site 10 XRF query has an archived 46-row result. Source-derived expected result tuples also specify the complete 48-observation genus profile for run `ERR16061083`, 360-row environment–taxon join, 70 site coordinates and six PCR no-template control roles. The release validator compares full result multisets, preserving duplicate rows and requiring exact identifiers. Numeric comparisons use absolute and relative tolerances of 10^{-12}; coordinate comparisons use an absolute tolerance of 10^{-10} degrees.

The query comparisons test the links between observations and their specimen, process and site context.

For KG v3.0.1, release validation verifies all 16 input-module hashes and parses the complete 45,707,660-quad export, checking every graph context. The terminology validation checks the term axioms and one definition per annotation property for each of 342 project terms. Public-release checks verify equivalent ontology content in RDF/XML, Turtle, JSON-LD and N-Triples, resolution of the four reference imports, manifest integrity, the live asserted-graph count and the deposited artifacts. Section 3 identifies the validation evidence and its input versions. Structural tests evaluate the scientific modules separately.

Table 8 Competency questions defining the retrieval workload.

ID	Competency question
CQ1	What are the XRF analyte concentrations measured at a given sampling site?
CQ2	Which sampling sites are recorded, and what are their GPS coordinates?
CQ3	Which biome and environmental features are assigned to each sampling site?
CQ4	What are the instrument-reported Light Elements values across the XRF measurements?
CQ5	Which DNA extraction records (extract, concentration, protocol) exist for a given sample?
CQ6	What is the full provenance chain from a soil sample to its sequencing run?
CQ7	What are the complete genus-level read counts and relative read abundance for run `ERR16061083`?

Listing 8 retrieves XRF measurements for a specific site, requiring traversal of the measurement process, quality, and value entities.

Listing 8 SPARQL query to retrieve field XRF measurements for Site 10. The query starts from the field-XRF measuring processes whose target is the site and walks to the individual analyte concentrations.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?processLabel ?siteLabel ?analyte ?concentration
WHERE {
  ?process a rak:0000025 ;
           rdfs:label ?processLabel ;
           sio:000291 ?site ;
           sio:000229 ?value .
  FILTER(CONTAINS(LCASE(STR(?processLabel)), "field xrf"))
  ?site rdfs:label ?siteLabel .
  FILTER(?siteLabel = "Site 10")
  ?value sio:000215 ?quality ;
         rak:2000012 ?concentration .
  ?quality a ?qualityClass .
  ?qualityClass rdfs:label ?analyte .
  FILTER(?qualityClass != rak:0000032)
}
```

```
ORDER BY ?processLabel ?analyte
```

The query selects field-XRF processes targeting Site 10, then retrieves each output and its analyte quality. It returns one row per process/analyte combination, excluding the aggregate Light Elements channel. The concentration units are undocumented.

The query returned 46 analyte-value rows for Site 10: 24 from Test 5847 and 22 from Test 5848. Validation compares the returned identifiers and values with the expected observations from the site and XRF data.

Listing 9 retrieves the complete genus profile for run `ERR16061083`, joining across bioinformatic workflows, FASTQ datasets, and taxon observations.

Listing 9 Complete genus-level profile for run ERR16061083, ordered by descending read count and lineage.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?proc ?fastq ?runLabel ?lineage ?taxon ?taxonLabel
       ?count ?relativeAbundance
WHERE {
  VALUES ?runLabel { "FASTQ dataset for ERR16061083" }
  ?proc a rak:0000071 ;
        sio:000230 ?fastq ;
        sio:000229 ?dsAbs, ?dsRel .
  ?fastq rdfs:label ?runLabel .
  ?dsAbs a rak:0000074 ; rdfs:label ?absLabel ;
         sio:000059 ?valAbs .
  ?dsRel a rak:0000075 ; rdfs:label ?relLabel ;
         sio:000059 ?valRel .
  FILTER(STRENDS(STR(?absLabel), "(Genus)"))
  FILTER(STRENDS(STR(?relLabel), "(Genus)"))
  ?valAbs rak:2000026 ?count ;
          rak:2000025 ?lineage ;
          sio:000215 ?qualAbs .
  ?valRel rak:2000020 ?relativeAbundance ;
          rak:2000025 ?lineage ;
          sio:000215 ?qualRel .
  ?qualAbs sio:000011 ?taxon, ?fastq .
  ?qualRel sio:000011 ?taxon, ?fastq .
  ?taxon rdfs:label ?taxonLabel .
  FILTER(?taxon != ?fastq)
}
ORDER BY ?runLabel DESC(xsd:double(?count)) ?lineage ?proc
```

The query binds both count and relative-abundance values to the same workflow, source FASTQ dataset, taxon and classifier lineage. Dataset labels encode rank through the suffix `(Genus)`. The query uses that emitted convention explicitly and returns all 48 observations for the named run. The source-derived expected table contains the complete profile, including each classifier lineage. The all-run query retrieves profiles from a locally loaded snapshot. That unrestricted export is a long-running operation. The environmental example below additionally binds field temperature and specimen collection to the same site visit.

Listing 10 SPARQL query to retrieve Pseudomonas relative abundance at sites with high temperature.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX pato: <http://purl.obolibrary.org/obo/PATO_>
PREFIX ncbitaxon: <http://purl.obolibrary.org/obo/NCBITaxon_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?visit ?visitLabel ?site ?siteLabel ?sample ?fastq
      ?temperatureProcess ?tempVal ?temp ?lineage ?relAbundance
WHERE {
  ?visit a rak:0000003 ; rdfs:label ?visitLabel ;
        sio:000291 ?site .
  ?site rdfs:label ?siteLabel .
  ?temperatureProcess sio:000068 ?visit ;
                      sio:000229 ?tempVal .
  ?tempVal rak:2000003 ?temp ; sio:000215 ?tempQual .
  ?tempQual a pato:0000146 ; sio:000011 ?site .
  FILTER(?temp > 35.0)
  ?sampling a rak:0000024 ; sio:000068 ?visit ;
           sio:000291 ?site ; sio:000229 ?collection .
  ?collection sio:000059 ?sample .
  ?ext sio:000230 ?sample ; sio:000229 ?dna .
  ?libPrep a rak:0000065 ; sio:000230 ?dna ;
          sio:000229 ?lib .
  ?seq a rak:0000066 ; sio:000230 ?lib ;
      sio:000229 ?fastq .
  ?bioinf a rak:0000071 ; sio:000230 ?fastq ;
         sio:000229 ?dsRel .
  ?dsRel a rak:0000075 ; rdfs:label ?datasetLabel ;
        sio:000059 ?valRel .
  FILTER(STRENDS(STR(?datasetLabel), "(Genus)"))
  ?valRel rak:2000020 ?relAbundance ;
         rak:2000025 ?lineage ; sio:000215 ?qualRel .
  ?qualRel sio:000011 ncbitaxon:286, ?fastq .
}
ORDER BY ?visit ?sample ?fastq ?tempVal ?lineage
```

Each result retains the collection visit, specimen, sequence dataset, temperature observation and genus lineage. A site's visits are joined through their process provenance. This provides an environmental temperature associated with the collection visit of each returned microbial profile.

Regression fixtures execute both queries on deliberately confusable inputs. The taxonomy fixture assigns the same genus different abundances in two profiles and adds a species-rank observation; the query returns exactly the two matching genus observations. The visit fixture gives one site two visits at 40 and 20°C; the temperature-filtered query returns only the specimen collected during the 40°C visit. These tests check the implemented joins; release-level validation additionally requires expected result tables for the complete modules.

4.3 Source reconciliation and measurement validation

The archived evidence workflow produced a specimen catalogue and reconciliation summaries for specimens, extracts, libraries, runs and community profiles. The 2,516 eligible source rows reconcile exactly to 2,516 generated sample individuals; the 1,647 extraction-eligible rows reconcile to 1,647 DNA extracts; and 1,242 resolved sequence rows reconcile to 1,242 amplicon libraries and 1,242 FASTQ datasets. A site-identity

check independently matches each Trip 1 numeric identifier 61–64 to a named catalogue individual by exact coordinates and rejects coordinate or IRI disagreement, accounting for all 36 corresponding source specimens.

Environmental validation accounts for 274 field observations and checks site identity, temperature, pressure and relative humidity. Values enter the curated environmental dataset only when they satisfy the measurement range and identity criteria. The XRF provenance audit reproduced all 13,390 reported Trips 1–4 cells under the maximum-positive rule and all 4,293 reported Trip 5 cells under the last-reported rule, with no mismatched reported cells, and verified that the canonical laboratory input comprises all 725 processed records. An independent chemical-identifier validation evaluated all 93 instrument channels against the pinned ChEBI release 247 and the checksummed PubChem snapshot of 23 July 2026 (Section 2.3.1); all rows passed the declared rules, with explicit nulls for the aggregate LE channel and unsupported oxides. The pH validation accounts for 1,168 specimen-assay entries and reproduces the five admission dispositions in Section 2.3.2. It checks specimen identity for every admitted measurement, including field replicate 2 for all 156 Trip 4 measurements.

4.4 MAG catalogue integrity

We checked the 5,438 refined MAG FASTAs against the representative inventory and the refined-bin CheckM2 report. Every genome's contig count and summed sequence length matched that report. The archive contains 2,485,945 contigs totalling 20,490,828,267 bp. We computed a SHA-256 checksum for each FASTA, then reread every archive member and verified its bytes against the per-genome manifest. The GTDB R232 classification tables passed their upstream checksum checks and contained exactly the same 5,438 genome identifiers. Library metadata reconciled all 551 read-bearing libraries between the paired-read inventory and the sample-context table.

The gene annotations were checked against the same sequences. Before annotating a genome, we verified its FASTA against the catalogue checksum. We accepted an annotation only when every eggNOG query was a protein that Bakta had predicted for that genome and each query occurred once. All 5,438 genomes passed, and the release step reread every annotation file against its recorded checksum. Bakta predicted 19,186,792 proteins, a median of 3,325 per genome. eggNOG-mapper annotated 84.0% of these proteins and assigned a KO to 44.4%, and every genome has at least one annotated protein. Across the catalogue, the annotations contain 11,386 distinct KOs, 9,703 Pfam domains and 141 CAZy families.

Each of the 551 coverage tables lists exactly the 5,438 catalogue genomes. All values are finite and non-negative, covered fractions are at most one, read counts are integers, and the reported length of each genome equals its total sequence length in the catalogue. The median library aligned 45.9% of its reads to the catalogue (range 3.9–83.8%). Seven libraries reached the limit of 200 million read pairs. Library F5Dr1, which yielded no assembly, contributed 616 read pairs.

4.5 ShEx schema validation

The archived ShEx run used the shapes described in Section 2.6.5. Its tractable modules passed their assigned shapes. The pH module was validated separately over all 712 measurement processes, values and acidity qualities and all 29 sessions, and the laboratory-control module passed its shape and scientific-invariant checks. Four negative fixtures each alter one constraint: missing site coordinates, an invalid value datatype, reversed value-to-quality direction, or an XRF process with neither input nor target. The released shapes reject all four fixtures. The multi-gigabyte taxonomy ABox remains outside this memory-bounded run; it is instead parsed in full by an independent Turtle parser and scanned by a streaming structural validator, which accounted for 44,528,482 triples in 5,953,690 subject blocks with zero structural violations. This checks full-file syntax and the generator's declared structural relationships. The separate listing validator checks printed Turtle against the modules available for that run. Taxonomy and external reference checks require the corresponding pinned payloads.

4.6 Logical consistency

ELK returned `true` for the tractable scientific-module subset loaded for validation, including the ecosystem, pH and laboratory-control modules. The checked subset excludes the taxonomy ABox and large reference modules. Its execution metadata identify the loaded axioms and import handling; class-hierarchy precomputation was omitted. The ontology uses `xsd:double` literals and constructs outside ELK's supported fragment[54]; the check therefore applies to the constructs processed by that configuration.

Validation of the separate finite-rule implementation uses synthetic examples for long hierarchies, transitive cycles, both project property-chain lengths, restriction and intersection classification, and PCR-control ancestry. The tests compare bounded, class-partitioned execution with an unbounded reference closure and check recovery from an interrupted run. Release-specific checks compare the complete control-role inventory with its source, retrieve consequences of the measurement chains and require the inferred graph to be disjoint from the asserted graph. Completion requires the zero-addition pass specified in Section 2.6.4, followed by the explicit clash diagnostics.

Label validation checks that each labelled project entity has exactly one label. The evidence identifies the input modules, their hashes and validator versions (Section 3).

4.7 Portal access

The portal (https://rubalkhali.science/) retrieves data through SPARQL queries and renders site and specimen views. Validation compares query results with source-derived expected values.

4.8 Identifiers and reuse metadata

Each named entity has a project-domain IRI. The manifest identifies payload files, and the data dictionary specifies curated fields, units and missing-value conventions.

RDF provenance links travel with the modules and support reuse through SIO, ENVO, ChEBI, PATO, UO and validated taxonomic identifiers[55].

The public endpoint provides read access to the deployed graph. The v3.0.1 Zenodo deposit fixes the released graph, module files and rebuild kit, with checksums, scoped licences and a version-specific DOI. Data Availability identifies the separate sequence-access requirement.

5 Usage notes

The portal at https://rubalkhali.science/ provides site pages and a SPARQL editor. The endpoint https://rubalkhali.science/sparql accepts SPARQL 1.1 queries. For a local, low-memory example, load the base, site and XRF modules listed in Table 7 and execute the Site 10 query in Listing 8. Compare its 46 returned observations with the distributed expected identifiers and values. Taxonomic queries also require the taxonomy module and its mapping dependencies. The named-run example retrieves the complete genus profile for `ERR16061083`; all-run exports require a locally loaded snapshot and a long-running join. The reproduction instructions accompanying the data provide retrieval, checksum, environment and execution details (Section 3).

5.1 Choosing observations for reuse

Use specimen, collection-visit and assay identifiers when joining observations. The primary repeated-campaign frame comprises sites 1–60; the source identifiers 61–64 refer to four named catalogue locations sampled in Trip 1. Several feature profiles share a source specimen identifier. Analyses should group these profiles by their documented specimen and processing relationships.

The control registry distinguishes assay, processing stage, extraction group and identity evidence. The Trip 5 candidate screen uses 17 extraction blanks linked to 217 biological profiles and identifies 351 ASVs. The canonical feature table retains these ASVs; the companion sensitivity analysis applies their removal to the linked profiles. Trip 4 has six separately denoised extraction blanks and three batches with unsequenced blanks, leaving complete or partial coverage gaps at 28 sites. Concentration measurements exist in the source records; any concentration-based screen requires links between specimens, extractions and concentration measurements. Four Trips 2–3 standards support genus-recovery evaluation. Trips 4–5 have extraction controls but no sequenced 16S positive-standard profiles.

Archived-soil pH describes specimens measured in $CaCl_2$ under the admission rules of Section 2.3.2. Coverage varies by campaign and compartment. All Trip 4 assays used archived field replicate 2. Their supported comparison with community profiles is at campaign, site and compartment level. The versioned specimen mapping links measurements to the confirmed specimens as described in Methods.

Field and laboratory XRF observations have separate acquisition provenance and undocumented concentration units. The released values support within-workflow descriptive comparisons; cross-workflow calibration requires documented units, paired aliquots and reference materials. Site 4 field readings include different substrates, so a site-level summary combines that local heterogeneity with instrument repeatability.

The PMA table contains three site-compartment groups at two campsites. Each group has one parent collection tube, from which three separate slurries supplied treated/untreated pairs. Descriptive paired summaries retain this preparation structure and compare dye treatment under matched dark and light exposure.

5.2 Versioning and supported operations

Use the versioned modules and their reference ontologies to reproduce queries. Updating an external ontology requires revalidating identifiers and mappings before regenerating the affected modules and query results.

KG v3.0.1 serves the asserted graph at https://rubalkhali.science/graph/asserted/3.0.1. This graph is the default dataset for portal and SPARQL queries. Explicit `FROM`, `FROM NAMED` and SPARQL protocol dataset parameters select the dataset for an individual request. The service evaluates stored assertions directly and rejects unavailable query modes.

The bundled finite-rule implementation supports the separate reasoning workflow described in Section 2.6.4. Publication of its output requires completion and validation of that workflow. Changes to released artifacts require a new semantic version.

A local copy of the deposited graph provides fixed inputs independently of the portal's deployed version. The rebuild kit verifies and loads the 16 distributed modules; generating the scientific modules uses the source data and code described in Section 3. Analyses requiring additional OWL consequences need a reasoner and validation workload suited to those constructs.

6 Data availability

The project uses ENA umbrella study `PRJEB104209`; its submitted amplicon run records are associated with `PRJEB106069`. ENA study `PRJEB125153` (secondary accession `ERP204894`) holds the shotgun data: 554 read runs from 551 soil BioSamples, 549 primary metagenome assemblies, the 12,434 refined bins as binned metagenomes, and 5,437 of the 5,438 representatives as MAGs with GCA accessions. ENA admits a genome as a MAG only below 10% contamination, so `48Dr2__self_SemiBin_25` is deposited as a binned metagenome. Both studies are scheduled for public release on 25 November 2026. The data repository lists the BioSample, run, assembly and genome accessions of every shotgun library, bin and MAG. The complete refined MAG sequence archive, its functional annotation tables, the coverage profiles and the accession tables are archived at Zenodo[56] and linked from https://rubalkhali.science/. The source and derived-data package is maintained at https://github.com/bio-ontology-research-group/empty-quarter-ecology-data. Its manifests distinguish included files, separately retrieved bulk payloads and unresolved upstream provenance. KG v3.0.1, its original modules and rebuild kit are archived at https://doi.org/10.5281/zenodo.23134168[53]. Project-authored RDF, ontology terms, definitions and dataset metadata in this release are licensed under CC BY 4.0. Imported vocabularies retain their respective licences, listed in the release's third-party attribution file; project software is MIT-licensed. Raw reads and other separately licensed materials retain their own access

and licensing conditions. The broader analytical tables and specimen and assay provenance are versioned in the reproducibility repository; the KG deposit contains the files enumerated in its manifest.

7 Code availability

The web application and ontology-development code are available at https://github.com/bio-ontology-research-group/empty-quarter. The data repository https://github.com/bio-ontology-research-group/empty-quarter-ecology-data contains the manuscript sources, RDF generators, metadata and Nextflow workflow. RDF generation uses Groovy scripts with OWL API v5.1.20[48]; the ELK validation wrapper uses OWL API v4.5.26. The repository supplies execution profiles, workflow tests, a pinned Conda recipe and a hash-locked Python environment.

Funding

Susana Martinez Arbas was supported by the Fonds National de la Recherche (FNR), Luxembourg (Project Ref. 18047750). No specific funding was received for this project.

Competing interests

The authors declare no competing interests.

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